D222 prefix-tuned summarizer: Methodology and Evidence

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Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

Ask whether learned task prefixes can steer a Phase 1 D222 model toward concise legal summaries with few trainable parameters. This adaptation changes the conditioning mechanism rather than inserting adapters into every transformation.

Architecture and mechanism

Trainable prefix key/value representations enter the attention pathway. The causal batch forms summary targets while the base model remains the source of legal language modeling knowledge.

Measured result

Scored submission	1031005
Codabench BERTScore F1	0.7710
Archived local metric	-1.6961 validation negative NLL; BERTScore not measured in this report

Codabench BERTScore F1 was 0.7710. One repeat upload failed, but the earlier finished submission is the scored evidence.

Limitations and next check

Prefix conditioning still depends on the parent model's document coverage and can miss facts beyond its context. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: d222_prefix.py, phase6_data.py, train_phase6.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.